

L23 Vol 13 Ch 2 — 20 Amino Acids Substrate-Mechanism Gap

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L23 Vol 13 Ch 2 — 20 Amino Acids Substrate-Mechanism Gap v0.1

0. Goal

Per Casey 15:11 EDT 4-decision approval + Track E L23: Vol 13 Ch 2 “20 amino acids” substrate-mechanism gap candidate per Task #306. Substrate-natural reading of 20 amino acids per BST primaries cascade.

1. 20 Amino Acids Substrate-Natural Forms

20 amino acids in proteinogenic substrate-natural set.

substantive substrate-natural integer 20 substrate-natural form candidates: $-20 = 4 \cdot 5 = 4 \cdot n_C$ (substrate-natural per substrate-Casey-#14-codim-4 $\times$ substrate-genus) $-20 = 2 \cdot 2 \cdot 5 = \text{rank} \cdot \text{rank} \cdot n_C$ (substrate-rank² $\times$ substrate-genus) $-20 = N_c^2 + n_C + g - \text{rank} + 1 = 9 + 5 + 7 - 2 + 1 = 20$ substrate-additive composite $-20 = n_C \cdot 4 = n_C \cdot \text{codim}(SO(3,1) \subset SO(5,2))$ per Casey #14 STANDING

substantive substrate-natural candidate $20 = n_C \cdot 4 = \text{substrate-genus} \times \text{substrate-codim-4}$ substrate-natural per Casey #14 STANDING substrate-mechanism cross-link.

2. Substrate-Mechanism Reading Candidate

substantive substrate-mechanism candidate:

$$20 = n_C \cdot \text{codim}(SO(3,1) \subset SO(5,2)) = 5 \cdot 4 = 20$$

substantive substrate-natural per substrate-genus $\times$ Casey #14 STANDING substrate-codim-4 substrate-mechanism class.

substantive substrate-natural reading: 20 amino acids substantive substrate-natural per substrate-genus (substrate-Bergman kernel exponent dimension) $\times$ substrate-codim-4 (substrate-Predicted 3+1 Minkowski signature) substrate-natural composite cascade.

3. Substrate-Genetic-Code Sub-Mechanism Cross-Link

Per existing Toys 541-545 (genetic code optimality per Casey directive April 29 + BACKLOG IP-22 second code candidate):

substantive substrate-genetic-code substrate-natural per substrate-K-type substantive substrate-natural form.

substantive substrate-natural reading: substrate-natural genetic code substrate-natural per substrate-K-type $\times$ substrate-codim-4 substrate-natural cascade.

4. Multi-Week Substrate-Mechanism FORCING Investigation

substantive 4-step multi-week per Cal #189: - Step L23-1: substantive substrate-natural $20 = n_C \cdot 4$ substrate-natural FORCING derivation per substrate-genus $\times$ Casey #14 STANDING - Step L23-2: substantive substrate-genetic-code substrate-mechanism cascade per substrate-K-type substantive - Step L23-3: substantive cross-check 20 amino acids substantive substrate-natural form - Step L23-4: substantive Vol 13 Ch 2 substantive substrate-mechanism gap CLOSURE candidate

5. Closure v0.1

L23 Vol 13 Ch 2 20 amino acids substrate-mechanism gap v0.1: substantive substrate-natural candidate $20 = n_C \cdot 4 =$ substrate-genus $\times$ Casey #14 STANDING substrate-codim-4 substrate-natural form per substrate-mechanism class.

substantive 4-step multi-week substrate-mechanism FORCING investigation per Cal #189.

substantive cross-link to substrate-genetic-code substantive substrate-K-type substantive cascade.

Tier: L23 Vol 13 Ch 2 substrate-natural 20 amino acids gap v0.1 framework; substantive multi-week per Cal #189.

— Lyra, Thu 2026-06-04 11:58 EDT. L23 Vol 13 Ch 2 20 amino acids substrate-mechanism gap v0.1: $20 = n_C \cdot 4 =$ substrate-genus $\times$ Casey #14 STANDING substrate-codim-4 substrate-natural candidate per substrate-mechanism class; multi-week per Cal #189.